Haonan LIU

Ph.D., Assistant Professor

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Profile

Haonan Liu is an Assistant Professor in the Department of Chemical Science and Engineering at the Institute of Science Tokyo. His research focuses on polymer analysis, particularly high-frequency dielectric properties, nanomechanics, high-pressure optical properties, and spectroscopic characterization of molecular dynamics.

Research Interests

Polymer dielectric materials for high-frequency applications (GHz-THz); low-loss polyimides; molecular dynamics-dielectric correlations; terahertz spectroscopy; atomic force microscopy and polymer nanomechanics.

Appointments

Assistant Professor, Nakajima Laboratory, Institute of Science Tokyo	Apr. 2026-Present
Assistant Professor, Ando Laboratory, Institute of Science Tokyo	Oct. 2024-Mar. 2026
Assistant Professor, Ando Laboratory, Tokyo Institute of Technology	Apr. 2023-Sept. 2024
Postdoctoral Researcher, Nakajima Laboratory, Tokyo Institute of Technology	Apr. 2022-Mar. 2023

Education

Dr. Eng., Chemical Science and Engineering, Tokyo Institute of Technology — Mar. 2022
Thesis: Morphological and nanomechanical characterization of block copolymer-based thermoplastic elastomers

M. Eng., Chemical Science and Engineering, Tokyo Institute of Technology — Mar. 2019
Thesis: Synthesis, structure, and physical properties of tripodal triptycene-appended vinylic polymers and copolymers

B. Eng., Chemical Engineering and Technology, Zhejiang University — June 2016
Thesis: Thermoresponsive gels with crystallizable side chains

Academic Service

Research Group on Computational Polymer Science, Organizing Committee	2026-Present
Japan Polyimide & Aromatic Polymers Conference, Organizing Committee	2023-Present

Selected Publications

- Sawada, R.; Liu, H.; Yanagimoto, S.; Yanagimoto, Y.; Ando, S. *Extremely Low Humidity and Frequency Dependence of Dielectric Properties of Highly Fluorinated Polyimides in the 10-330 GHz Range. ACS Applied Polymer Materials*, 7, 14371-14381 (2025).
- Sawada, R.; Liu, H.; Ando, S. *Vibrational Spectroscopic Analysis of Water Absorption in Polyimides and the Correlation with Dielectric Properties at 10 GHz. The Journal of Physical Chemistry B*, 129, 10928-10943 (2025).
- Tsurusaki, Y.; Sawada, R.; Liu, H.; Ando, S. *Optical, Dielectric, and Thermal Properties of Bio-Based Polyimides Derived from An Isosorbide-Containing Diamine. Macromolecular Rapid Communications*, 46, 2401113 (2025).
- Liu, H.; Sawada, R.; Yanagimoto, S.; Yanagimoto, Y.; Ando, S. *Frequency-Dependent Dielectric Properties of Aromatic Polyimides in the 25-330 GHz Range. Applied Physics Letters*, 124, 232903 (2024).
- Sawada, R.; Yajima, K.; Takao, A.; Liu, H.; Ando, S. *Thermal, Optical, and Dielectric Properties of Bio-based Polyimides Derived from an Isosorbide-containing Dianhydride and Diamines with Long Alkyl Chains. Journal of Photopolymer Science and Technology*, 37, 507-516 (2024).

Selected International Presentations

- Liu, H.; Sawada, R.; Tsurusaki, Y.; Ando, S. *Broadband (10-330 GHz) Dielectric Characterization of Polyimides for High-Frequency Communication Applications.* International Polyimide Conference (IPIC2025), Shanghai, China, 9-12 July 2025.

- Liu, H.; Sawada, R.; Ando, S. *Highly Fluorinated Polyimides Exhibiting Extremely Low Frequency and Humidity Dependence in High-Frequency Broadband (10-330 GHz) Dielectric Properties.* 12th World Congress of Chemical Engineering (WCCE12), Beijing, China, 13-18 July 2025.
- Liu, H.; Sawada, R.; Ando, S. *High-Frequency Broadband (10-330 GHz) Dielectric Properties of Polyimides Applicable to 5G/6G Wireless Communications.* APME2025, Catania, Italy, 4-8 May 2025.
- Liu, H.; Sawada, R.; Yanagimoto, S.; Yanagimoto, Y.; Ando, S. *Dielectric Properties of Various Polyimides in the mmWave Range.* Polycondensation 2024, Lyon, France, 15-18 Sept. 2024.
- Liu, H.; Sawada, R.; Yanagimoto, S.; Yanagimoto, Y.; Ando, S. *Dielectric Dispersion of Polyimides in the Millimeter Wave Frequency (25-330 GHz) Range.* International Polyimide Conference (IPIC2024), Beijing, China, 18-21 June 2024.